Supplementary Material for *Large-scale traffic flow reconstruction from heterogeneous mobility sensors*

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This supplementary material documents the meaning of the data layers, model variables, data-assimilation logic, validation metrics and selected-road illustrations supporting the main manuscript. Numerical choices are reported with their interpretation: some follow from the temporal resolution of the observations, some stabilize the simulator, and one (the 40 km separation of the example links) only serves the figures. The scripts that produce the revised numbers are listed in Section S7.

S1 Data Layers

The reconstruction combines four observational layers and one derived road support. Their roles are distinct and are not interchangeable (Table S1). The mobile-phone layer is the only one that describes regional demand at broad areal scale, but it is too coarse to say which road was used. The road graph provides the missing physical support for route assignment. Coils provide local counts, but only where a sensor exists. Viasat is spatially richer than coils, but it is a probe sample rather than a total-count system. The reconstructed link-flow table combines these partial views under the model assumptions described below.

Table S1: Meaning and role of the data layers.

Layer	Spatial/temporal support	Meaning in the workflow
Mobile-phone OD activity	Municipality pairs, hourly profiles	Mobility-activity demand. It defines where trips are generated and how demand changes over the day, but it does not identify the road sequence used by each trip.
Road graph	Directed polylines and nodes	Physical support on which demand is assigned. It stores connectivity, directionality, segment length and turn constraints.
MTS inductive loop detectors (coils)	Fixed points, 15-min counts	Local direct traffic measurements. They anchor the simulation and drive the data assimilation where observed counts are available.
Viasat GPS telemetry	Matched road links, hourly counts	Independent validation layer. Because it samples only part of the vehicle population, it is compared with simulation after estimating link-specific penetration.
Derived link-flow tables	Directed road links, hourly counts	Simulation output after optional data assimilation. These tables are the link-level reconstruction used for maps and validation.

S1.1 Demand Structure

The weekday and weekend OD matrices are highly concentrated. This motivates using the observed spatial demand instead of injecting uniform trips across the region. The total-flow values in Table S2

are the activity weights entering the demand generator after aggregation over the selected January 2020 period. They are relative mobility mass, not vehicle counts. Positive cells count municipality pairs with non-zero activity. The Gini coefficient and top-share columns quantify concentration: values close to 0.8 and a top-1% share above 0.4 mean that a small number of OD relations dominate the signal. The model therefore uses the observed high-demand centres explicitly and assigns weakly observed endpoints to the random classes.

Table S2: Structure of the weekday and weekend (Saturday–Sunday) OD matrices.

Period	Total flow	Positive cells	Gini	Top 10% share	Top 1% share
Weekdays	2,665,186	87,212	0.788	0.708	0.448
Weekends	2,198,516	87,447	0.777	0.690	0.432

The ranking in Table S3 is used only to describe the heterogeneity of municipal activity and to identify the dominant urban centres. The last-ranked municipalities have mean activity equal to one because they are at the residual tail of the aggregated demand table. They are kept in the descriptive statistics, but the simulator does not assume that such low-activity entries carry enough information to define precise origin–destination routes on their own.

Table S3: Municipality activity ranking used to define high-demand centres.

Highest activity			Lowest activity		
Rank	Municipality	Mean points	Rank	Municipality	Mean points
1	Bologna	345,267	519	Montecalvo in Foglia	1
2	Modena	176,166	520	Porto San Giorgio	1
3	Parma	158,440	521	Trento	1
4	Ravenna	147,772	522	Montepulciano	1
5	Rimini	144,696	523	Vescovato	1
6	Reggio nell'Emilia	139,849	524	Tortona	1
7	Ferrara	103,067	525	Aviano	1
8	Piacenza	92,678	526	Malborghetto Valbruna	1
9	Forlì	86,078	527	Bolzano	1
10	Cesena	78,582	528	Grottammare	1

S1.2 Sensor and Telemetry Coverage

Table S4 reports data volume in terms of records, variables and physical interpretation. The 528 municipalities define the areal units of the demand layer. The 292 traffic stations are sparse compared with the road graph, so the assimilation is local and directional. The 5,760 rows of the 2020 coil table correspond to 60 days at 15-min resolution; each day contributes 96 samples per direction. The Viasat campaigns contain hundreds of thousands of hourly link observations, but the 4,186 matched road identifiers are still only a subset of the regional graph. This justifies using Viasat for temporal-profile validation and not as a complete observation layer.

S2 Simulation Runs and Source Classes

The final validation archive contains sixteen daily simulations, eight in February 2020 and eight in June–July 2021. The 5 July 2021 run is excluded from every primary validation result, leaving fifteen evaluated dates. The source-class totals below describe all sixteen archived configurations and are provenance only. The two periods were selected because independent Viasat validation campaigns are available for those dates. Viasat is not used to configure or assimilate the simulations; it provides the held-out probe layer

Table S4: Observed data volumes used by the paper workflow.

Layer	Records	Variables	Interpretation
Municipality activity table	528	5	Ranked source and destination activity over municipalities.
Coil registry	292	5	Geographical catalogue of traffic-count stations.
Coil-to-road matching	292	9	Association between each traffic station and a graph polyline.
February–March 2020 coil table	5,760	496	Fifteen-minute directional traffic counts used for the coil-282 comparison and data-assimilation support.
Viasat 2020 campaign	803,904	10	Hourly GPS link-flow observations over 4,186 matched road identifiers.
Viasat 2021 campaign	703,416	10	Hourly GPS link-flow observations over the same matched road universe.

against which simulated link profiles are compared. The two weeks do not span the seasonal variability of traffic.

Every run uses a 10 s integration time step and writes hourly outputs. No numerical time-step convergence study has been completed for this assessment. Hourly output is used because both the mobile-phone demand profiles and the validation figures are expressed at hourly scale; it also reduces noise relative to 15-min sensor counts while preserving morning, midday and evening traffic structure. The source count changes by day because the OD generator activates only demand components present in the selected daily schedule.

Table S5: Archived daily simulation configurations, summarized by meaning rather than file name.

Date	Sources	Output support
2020-02-10	758	Hourly directed links
2020-02-11	740	Hourly directed links
2020-02-12	762	Hourly directed links
2020-02-13	736	Hourly directed links
2020-02-14	722	Hourly directed links
2020-02-15	758	Hourly directed links
2020-02-16	980	Hourly directed links
2020-02-17	758	Hourly directed links
2021-06-28	763	Hourly directed links
2021-06-29	744	Hourly directed links
2021-06-30	766	Hourly directed links
2021-07-01	740	Hourly directed links
2021-07-02	726	Hourly directed links
2021-07-03	763	Hourly directed links
2021-07-04	983	Hourly directed links
2021-07-05	763	Hourly directed links

The inherited source-count interpretation suggests that the larger counts on Sundays arise from the way residual weekend mobility is represented: when the observed OD structure is less dominated by commuting pairs, more source components are needed to preserve regional coverage. This does not mean

that Sunday traffic is necessarily larger than weekday traffic; it means that the demand generator uses a larger set of active source definitions to represent a more diffuse mobility pattern.

Table S6: Source-class counts across the archived simulations.

Class	Min/day	Max/day	Total	Endpoint logic
Fixed municipality OD	563	804	9,843	Observed source and observed destination.
Fixed province OD	55	78	1,003	Province-level source and destination.
Source-random	50	50	800	Observed source, sampled destination.
Random-destination	50	50	800	Sampled source, observed destination.
Random-random	1	1	16	Both endpoints sampled from the modelled region.

The fixed municipality OD class dominates because it is the part of the demand where both endpoints are directly observed. The two one-sided random classes are kept at 50 sources per day so that residual mobility can enter and leave the main observed centres without overwhelming the empirical OD matrix. The single random–random source is a low-intensity background component: it prevents the model from suppressing trips with neither endpoint in the dominant observed set, but it is deliberately small so that it cannot create an artificial uniform traffic field.

S3 `minimocas` Model Details

`minimocas` is an agent-based road simulator. Vehicles are represented by agents that move on directed polylines, choose a next node at junctions and accumulate link-level counts. The model state is summarized in Table S7. Agents are individual vehicles, without lane changing or car-following dynamics; the output of interest is the directional link-by-time flow field.

Table S7: Main vehicle-agent state variables.

State variable	Type/unit	Role
Identifier	integer	Distinguishes each simulated vehicle.
Status	categorical	Tracks whether the vehicle is active, completed, waiting, queued or interacting with transport logic.
Current road link	polyline id	Link occupied by the vehicle at the current time step.
Position on link	m	Longitudinal coordinate along the occupied link.
Signed speed	m s^{-1}	Magnitude gives speed; sign encodes direction.
Destination	node id	Current graph node pursued by the routing rule.
Previous and next node	node ids	Junction state used to move between links.
Route-deviation parameter	probability	Controls whether the vehicle follows or departs from the precomputed best path.
Distance and time counters	m, s	Accumulate travelled distance, trip time and lifetime.

Best paths are computed on a weighted graph with a Dijkstra-style engine. In the archived validation runs the routing weight reduces to polyline length, while forbidden turns are still excluded. The movement

update also contains a density-dependent speed rule:

$$v_p^{\text{target}} = \begin{cases} v_{\text{crit}}, & \rho_p > \rho_{\text{crit}}, \\ (1 - \rho_p/\rho_{\text{crit}})v_p, & \rho_p \le \rho_{\text{crit}}, \end{cases} \qquad v_p^{\text{target}} \ge v_{\text{crit}}. \tag{1}$$

The validation configuration uses $\rho_{\text{crit}} = 3/4.20$ m^{-1} and $v_{\text{crit}} = 10/3.6$ m s^{-1}.

The critical density corresponds to a compact but still interpretable spacing: roughly three vehicles over 4.2 m in the internal density convention used by the simulator. Its role is not to calibrate a detailed fundamental diagram; it prevents unrealistically high speeds when many agents occupy the same link. The critical speed is 10 km h^{-1}, converted to SI units, and acts as a lower operational speed in congested conditions. The rule should therefore be read as a stabilizing traffic-flow constraint rather than as a fitted car-following law.

Table S8: Numerical parameters and switches in the archived validation runs.

Parameter	Value	Role
Integration step	10 s	Numerical simulation step.
Output sampling	3600 s	Temporal resolution of saved link counts.
Source activation	30 s	Interval at which new vehicles are generated.
State grid cell	5000 m	Spatial scale of gridded state output.
Cartographic index cell	10000 m	Spatial index used for graph lookup.
Length routing weight	1.0	Makes routing cost proportional to length.
Speed routing weight	0.0	Disables speed-dependent routing in archived runs.
Initial speed range	15–25 m s^{-1}	Sampling range for generated vehicles.
Random-endpoint time budget	5600 s	Maximum time field used by random source classes.
Unique-link output	active	Saves the directional link counts used in the paper.

The values in Table S8 reflect a compromise between temporal fidelity and reproducibility. A 10 s integration step is small relative to the one-hour saved output and keeps route progression smooth. The 3600 s sampling interval is chosen because the OD demand and Viasat validation are analysed hourly; a finer output would mostly add sensor noise. The 30 s source-activation interval spreads injected agents over each hour instead of creating a single burst at the hour boundary. The 5000 m state grid and 10000 m cartographic index are lookup scales used to organize regional graph operations; they are much larger than individual road links, so they do not define the resolution of the final traffic field. Setting the length routing weight to 1.0 and the speed routing weight to 0.0 makes route choice depend on path length in the archived runs. This choice avoids mixing route assignment with uncertain speed calibration when the main objective is to validate link-flow profiles. The 15–25 m s^{-1} initial-speed range covers typical extra-urban speeds before density updates. The 5600 s random-endpoint time budget is a soft bound used when sampling residual trip endpoints: it allows trips longer than a one-hour commute but avoids generating unrealistic cross-region random journeys as the dominant residual class.

S4 Coil-based Data Assimilation

The data-assimilation step adjusts simulated unique-link counts using MTS coil residuals. Directional coil columns are mapped to cartographic `IN`/`OUT` directions. The previous revision identifies the 30 March 2023 correction code as the archived implementation and reports a machine-precision reproduction of one corrected table. This historical report has not been independently re-established across every current variant. In that version, gaps in an observed coil table are filled by linear interpolation in time, and when no observed table covers the simulated day (the 2021 runs) the weekday-specific seasonal average of each coil is used. The operational code now in the webserver fills missing values with seasonal weekday averages that combine the most recent year and earlier years with inverse standard-deviation weights.

For a coil-associated directed link i, let $u_i(h)$ be the unique-link count, $b_i(h)$ the simulated virtual-barrier count and $c_i(h)$ the measured coil count. The data-assimilation step estimates a barrier-to-unique

relation and a local relative residual,

$$r_i(h) = \frac{b_i(h)}{\max(u_i(h), 1)}, \qquad \Delta_i(h) = \frac{c_i(h) r_i(h) - b_i(h)}{\max(b_i(h), 1)}. \tag{2}$$

This residual becomes the additive increment $\delta_i(h) = u_i(h)\Delta_i(h)$ on the local directed link, so that the assimilated count equals $c_i(h)$ whenever $u_i, b_i \geq 1$, and is propagated to neighbouring links according to simulated turning shares. The ratio $r_i(h)$ is needed because a virtual barrier crossing and a unique-link occupancy count are not the same measurement. The former counts passages through a point-like section, whereas the latter counts distinct agents present on the whole directed polyline during the output interval. Using $r_i(h)$ converts the coil residual into the same scale as the simulated barrier before it is applied to the unique-link table. The denominator thresholds $\max(u_i(h), 1)$ and $\max(b_i(h), 1)$ prevent division by zero in night-time or low-flow hours; they do not add vehicles to the output, but only regularize the residual calculation.

Table S9: Stopping and exclusion rules in recursive data assimilation.

Rule	Implementation
Minimum propagated increment	Stop if the absolute assimilation increment is below one vehicle.
Maximum graph depth	Archived runs: stop once more than three links have been visited from the same coil (the counter is shared by all branches). Webserver version: five links.
Other sensor links	Archived runs: stop at a node as soon as one of its outgoing links is a coil polyline. Webserver version: skip only that link.
Forbidden turns	Do not propagate through transitions forbidden in the road graph.
Closed or patched roads	Do not propagate through excluded polylines.
Negative counts	Clip negative assimilated counts to zero before delivery.

The one-vehicle stopping rule in Table S9 is a discreteness threshold: below one vehicle an increment has no meaning in an hourly count table. The depth limit keeps the correction local, so that a single coil cannot reshape a large part of the regional network, and stopping at other sensor links leaves each observed location under the control of its own counter. Two features of the implementation should be kept in mind when reading the results. In both versions the increment passed to the second and later outgoing branches of a node is multiplied by the weights of the branches already processed, so those branches receive less than their turning share. Furthermore, the MTS tables store UTC timestamps, which both versions compare with the local clock of the simulation without conversion; the effect of this offset on the validation is quantified in the main text.

S4.1 Leave-one-out test

The archived leave-one-out table contains 1,202 directional records over 610 coil-days, with 18 expected directions missing. Its runner skips absent corrected directions. A new one-day pilot observed such a loss on a terminal link; complete directional-support validation remains open. The leave-one-out test uses the assimilation code of the archived runs and the raw simulations of 10–17 February 2020. For every day and every coil with all 96 quarter-hours observed in both directions, the coil is removed from the list of assimilated coils and from the list of polylines at which the propagation stops, so that increments from neighbouring coils can reach its link. The assimilation is then rerun and the hourly counts $\tilde{u}(h)$ on the held-out link are compared with the hourly coil counts $c(h)$, in the same clock convention used by the assimilation. The error of a coil is $\sum |\tilde{u}(h) - c(h)| / \sum c(h)$ over hours, directions and days. Table S10 and Fig. S1 summarize the results. When the coil is kept in the assimilation the same error is 1.2%, the residual coming from hours with no simulated vehicle or barrier crossing on the link, where the increment does not reproduce $c(h)$. Of the 76 coils reached by the propagation, the median error goes from 49.4% to 44.7%.

Coil 282 is used in the main manuscript because both directions have complete 15-min coverage for the selected day. The daily sums are similar but not identical, as expected for a road with different dominant

Table S10: Leave-one-out test of the coil assimilation, 10–17 February 2020. For each coil with complete data on a given day, the coil is removed from the assimilation and from the set of polylines that stop the propagation; the error is $\sum_h |u(h) - c(h)| / \sum_h c(h)$ on the held-out coil link, summed over directions and days.

	Raw simulation	After assimilation
Pooled relative error [%]	59.8	56.7
Median over coils [%]	46.2	44.3
Interquartile range [%]	33.2–62.2	32.3–62.0
Coils tested		149
Coil-days		610
Coils reached by the propagation [%]		51
Coils improved / worsened [%]		39 / 13

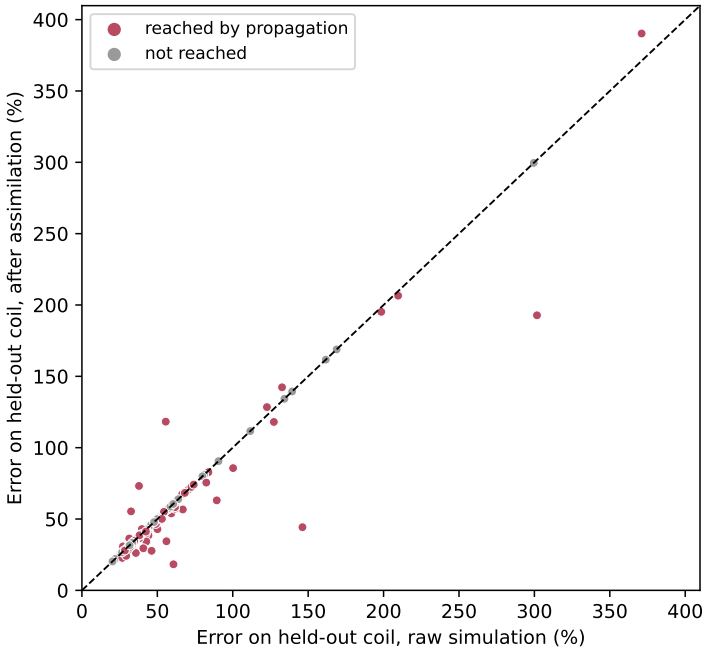


Figure S1: Leave-one-out test: error on each held-out coil link before (raw simulation) and after assimilating the other coils, 10–17 February 2020. Red: coils whose link receives an increment from the propagation; grey: coils not reached.

Table S11: Observed coil 282 data on Monday 10 February 2020 (local time).

Direction	Fifteen-minute samples	Daily sum	Peak hourly count	Peak hour
0	96	11,538	1,166	17:00
1	96	10,397	1,143	07:00

directions in the morning and in the afternoon. The opposite peak hours, 17:00 in direction 0 and 07:00 in direction 1, are consistent with a road segment that carries different dominant directions in the morning and afternoon.

S5 Validation Tables

The GPS-to-simulation ratio is defined in the main text. It depends on the model denominator and must not be interpreted as an independently observed fleet penetration. The archived class-stratified validation table below is reproduced as provenance; the three-link 2021 upper class is displayed explicitly and should not be overinterpreted. No significance test treating neighbouring links as independent is used in the new analysis.

Table S12: Profile error of the assimilated simulation against penetration-normalized Viasat profiles on all usable matched links, by campaign and penetration class η. MAE in percentage points (Eq. (3)); ε is the median relative error (Eq. (4)); r is the median Pearson correlation of the hourly shares. The 10^{-5} class (three links in 2020, two in 2021) is included in the totals only; the three-link upper class in 2021 is displayed for provenance.

Campaign	Class	Links	Median MAE	IQR	10th–90th pct.	ε [%]	r
2020	all	848	1.34	1.05–2.01	0.87–3.28	32	0.82
	10^{-1}	17	0.94	0.91–0.97	0.79–1.12	23	0.91
	10^{-2}	514	1.13	0.96–1.34	0.83–1.61	27	0.87
	10^{-3}	275	2.13	1.66–2.82	1.43–3.65	51	0.66
	10^{-4}	39	5.11	4.63–6.77	3.77–7.50	123	0.36
2021	all	845	1.49	1.15–2.12	0.98–3.48	36	0.76
	10^{-1}	3	1.13	–	–	27	0.85
	10^{-2}	518	1.24	1.03–1.51	0.89–1.86	30	0.83
	10^{-3}	289	2.18	1.73–3.06	1.43–3.89	52	0.64
	10^{-4}	33	5.75	4.86–6.82	4.67–7.64	138	0.27

The candidate set for the example links is the upper class of a three-class k-means partition of the standardized $\log_{10}\eta_i$, which in 2020 contains the 487 links with $\eta_i \geq 0.40\%$. Before clustering, each simulated profile is divided by its daily total, so that Ward linkage acts on the 24 hourly shares and not on volumes. On raw counts the Ward distance is dominated by volume: the three raw-count clusters used in the earlier version of this analysis (74, 112 and 301 links) are disjoint bands of daily traffic, 1,001–6,384, 6,597–15,446 and 14,638–55,856 simulated vehicles per day. The partition on shares has an adjusted Rand index of 0.18 with the raw-count partition, so the two groupings are largely different. The dendrogram is cut into three clusters, at Ward distance 0.15; three is also the number of example links that fit in one validation figure. Figure S2 shows the mean hourly shares of the three subclusters.

The selection of the example links in Table S14 follows three rules. The three roads belong to different subclusters. Polyline 104671 is fixed because its position near Forlimpopoli gives it a direct interpretation. In the other two subclusters, among the links with at least 300 simulated vehicles per day and 50 Viasat probes in the week (all 487 candidates), we take the pair with the lowest total MAE whose pairwise distances, including the distance to 104671, are at least 40 km. The two selected links are thus among the best of the candidate set, as the last column of Table S14 shows; the full distribution is in Table S12 and Fig. 5 of the main text.

S6 Additional validation and uncertainty

S6.1 Declared one-hour display shift

Figure S5 shows the pooled simulated profile translated forward by one hour as a visual diagnostic. The original profile is retained for comparison. This translation changes only the plotted time coordinate:

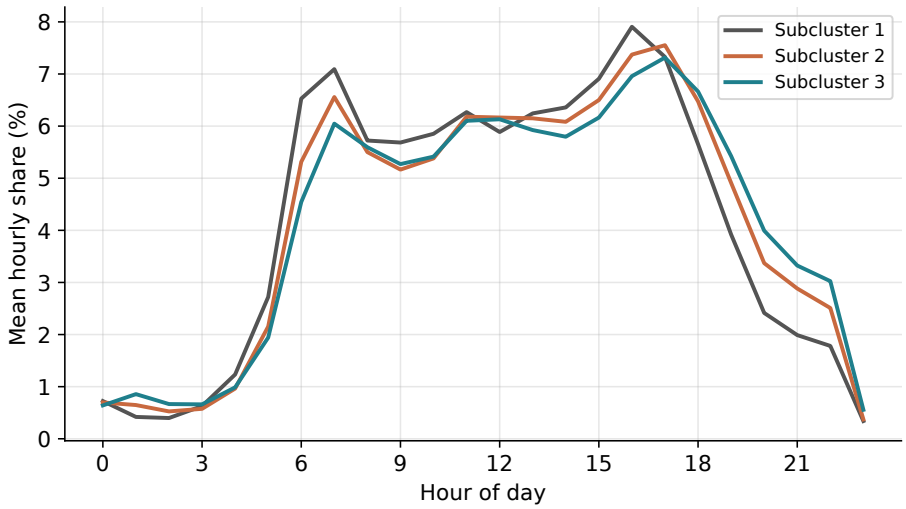


Figure S2: Mean normalized simulated profile of the three subclusters of Fig. S3 (51, 78 and 358 links).

Table S13: Candidate-filtering logic for the representative validation links.

Selection stage	Links	Purpose
Common simulated–Viasat links in 2020	930	Initial matched comparison set.
Usable hourly profiles	848	Non-zero simulated and GPS totals.
Upper k-means class of $\log_{10}\eta_i$	487	Candidates for subclustering, $\eta_i \geq 0.40\%$.
Displayed representative links	3	One illustrative link per temporal subcluster.

Table S14: Representative polylines of Fig. S4 in this supplement. The last column is the percentage of the 848 usable 2020 links with a lower MAE. Minimum pairwise separation: 45.0 km.

Subcluster	Polyline	MAE [pp]	ε [%]	Links with lower MAE [%]
1	104671	1.22	29	40
2	134419	0.81	19	5
3	64544	0.72	17	1

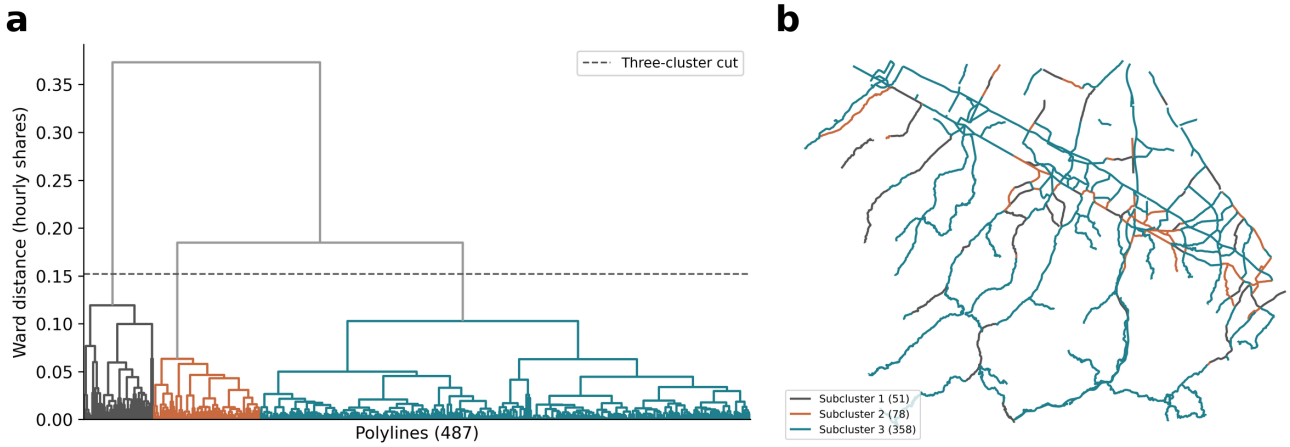


Figure S3: **Subclusters of the high-penetration links in 2020.** **(a)** Ward dendrogram of the normalized hourly profiles of the 487 candidate links; the dashed line marks the three-cluster cut. **(b)** Location of the three subclusters, with the number of links in parentheses.

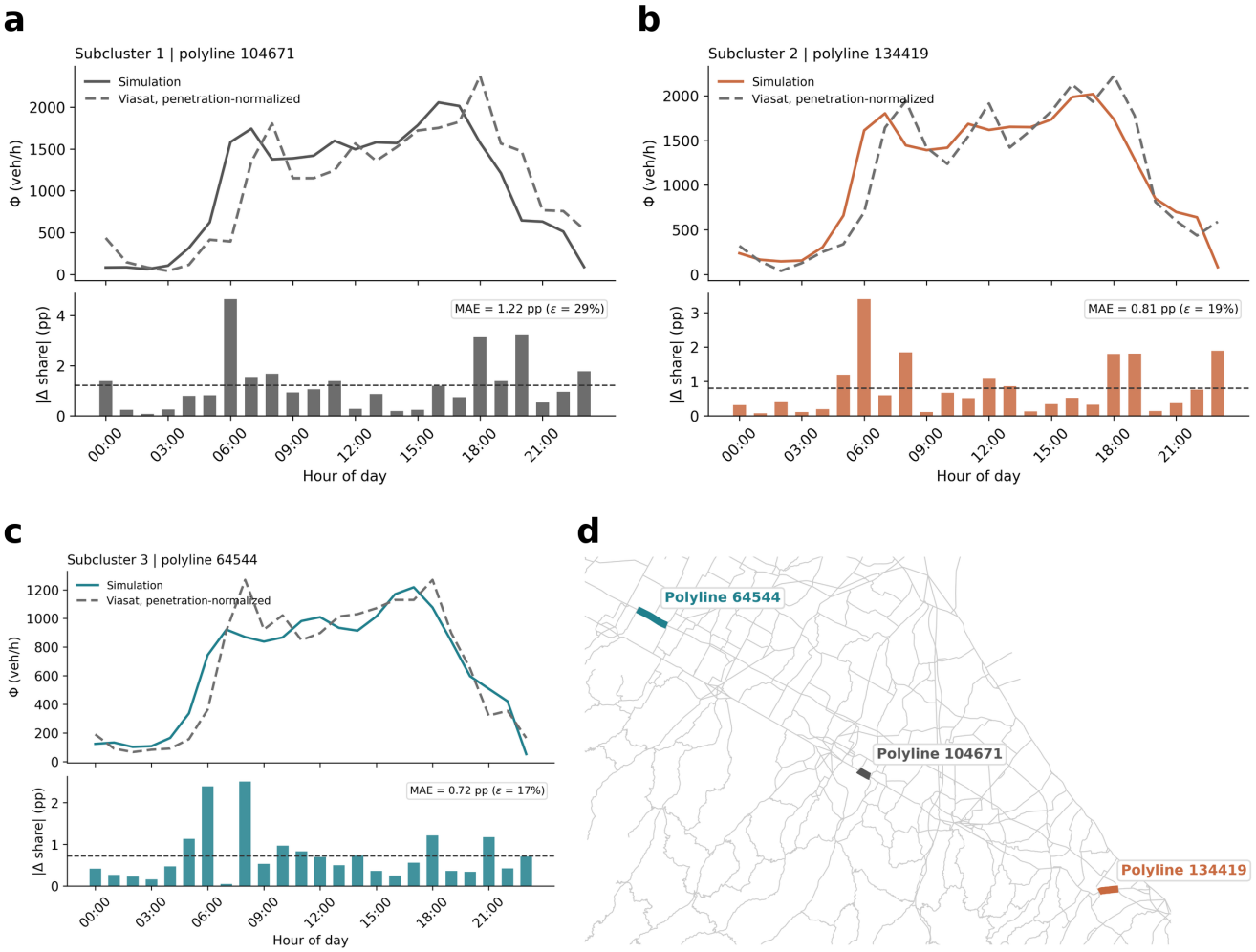


Figure S4: **Representative links, 10–17 February 2020.** **(a–c)** Mean simulated hourly flow and penetration-normalized Viasat profile (top) and hourly absolute share difference $|p_i(h) - q_i(h)|$ (bottom). **(a)** Subcluster 1, polyline 104671, MAE = 1.22 pp. **(b)** Subcluster 2, polyline 134419, MAE = 0.81 pp. **(c)** Subcluster 3, polyline 64544, MAE = 0.72 pp. **(d)** Location of the three polylines on the road graph, coloured as in Fig. S3.

the 23:00 value is displayed at 24:00 without circular wrapping. It does not correct the simulation or establish the cause of its lead. The primary validation uses the original time coordinates and unchanged data. Pooled GPS and simulated curves weight links by their own counts, as in the main diagnostics.

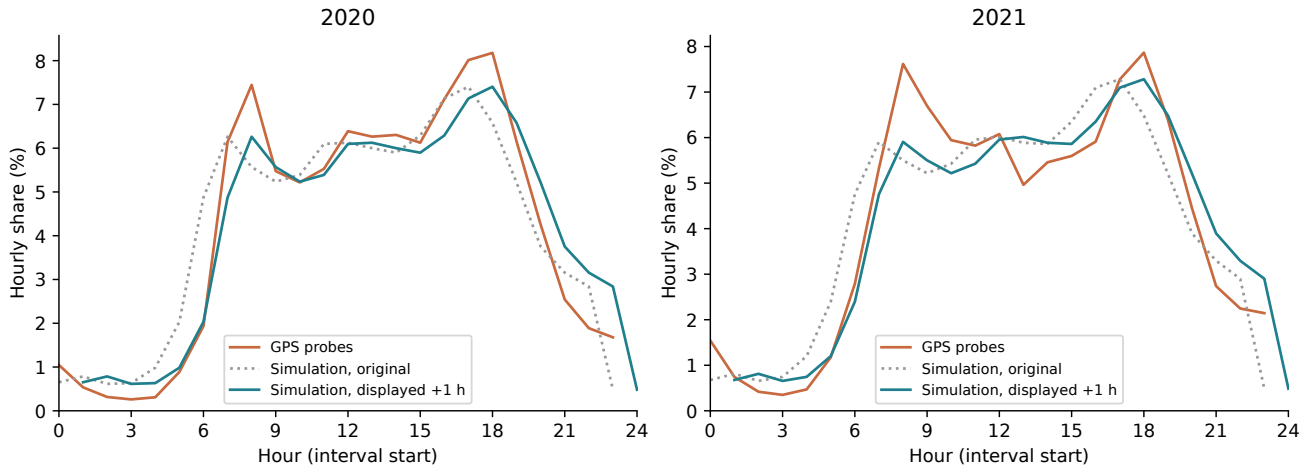


Figure S5: **Declared timing sensitivity.** Pooled GPS profiles, original simulated profiles and the same simulated profiles displayed one hour later. The +1 h translation is applied for this illustration only; it is not an inferred physical correction or an independent validation result.

The new analysis separates weekdays and weekends and stratifies by total probe count without dividing by a model volume. These groups describe the available archive, not independent traffic regimes or experimental replicates. Each weekend contains only two dates. Fewer observations increase uncertainty of a normalized hourly profile, so a weekend error increase cannot be assigned uniquely to the demand model.

Table S15: Profile error by total campaign probe count, independent of the simulated-volume denominator. Groups are descriptive; probe trajectories and neighbouring links are dependent.

Campaign	Probes	Links	Assimilated MAE	MTS MAE	Correlation
2020	1-49	120	3.78	3.76	0.44
2020	50-199	193	1.93	1.82	0.70
2020	200+	535	1.12	0.82	0.87
2021	1-49	125	3.89	3.87	0.37
2021	50-199	228	1.96	1.81	0.68
2021	200+	492	1.21	0.89	0.83

The updated 2021 mean MTS comparator excludes direction profiles containing all-missing hourly bins. The old group-by sum converted these bins to zeros. Its median error changes from 1.214265 to 1.215563 pp; all primary reconstruction, OD and correlation metrics reproduce the inherited computations within 10^{-9}. This difference does not reverse the baseline comparison.

The sensor-cluster bootstrap samples whole counters with replacement and retains each counter's available dates and directions. For 5,000 replicates and seed 20261027, the pooled error reduction is 3.0809 pp, with percentile interval 0.3937–7.0201 pp. The interval describes resampling of the observed sensor set; spatial dependence, implementation bias and uncertainty in demand are outside its scope. Leave-one-date-out reductions range from 2.9266 to 3.2550 pp. These are diagnostic checks on the historic UTC-as-local experiment.

S6.2 Exploratory sensor-coverage, clock and depth pilot

A separate experiment uses 10 February 2020, requiring complete observed data and usable graph-direction support. It excludes terminal-link coils because the historic correction can omit their held-out direction.

There are 56 eligible coils; a fixed random split (seed 20261027) withholds 11 and retains 45 in the training pool. Training subsets contain approximately 25%, 50% or all of that pool, with three nested-order repeats. The nominal shared recursion budgets are 1, 3 and 5. Each variant uses either the archived counter clock or local-time training observations. Every test target is local time, and test coils are absent both from correction inputs and propagation-stopping links. This gives 54 specified runs; the full-training repeats are identical, not independent replications.

At budget 3, the raw pooled error on the 11 test coils is 55.61999%; using all 45 training coils with local-time observations changes it to 55.61670%, a reduction of only 0.00329 percentage points (Table S16). This restricted one-day result shows almost no transfer to this fixed test set. It does not establish coverage insensitivity in general, compare directly with the historical LOO population, or select an optimal depth. The retained implementation still has sibling-weight and shared-budget defects. Broader sensor coverage, geographic holdouts and repaired output support remain necessary.

Table S16: Exploratory 10 February 2020 fixed-holdout pilot at shared depth budget 3. All test targets are local time. Mean and range across three nested-training repeats describe split sensitivity, not confidence intervals. At full training the three sensor subsets are identical.

Training-pool share	Input clock	Mean error (%)	Range
25%	archived	55.62	55.62–55.62
25%	local	55.62	55.62–55.62
50%	archived	55.62	55.62–55.62
50%	local	55.62	55.62–55.62
100%	archived	55.62	55.62–55.62
100%	local	55.62	55.62–55.62

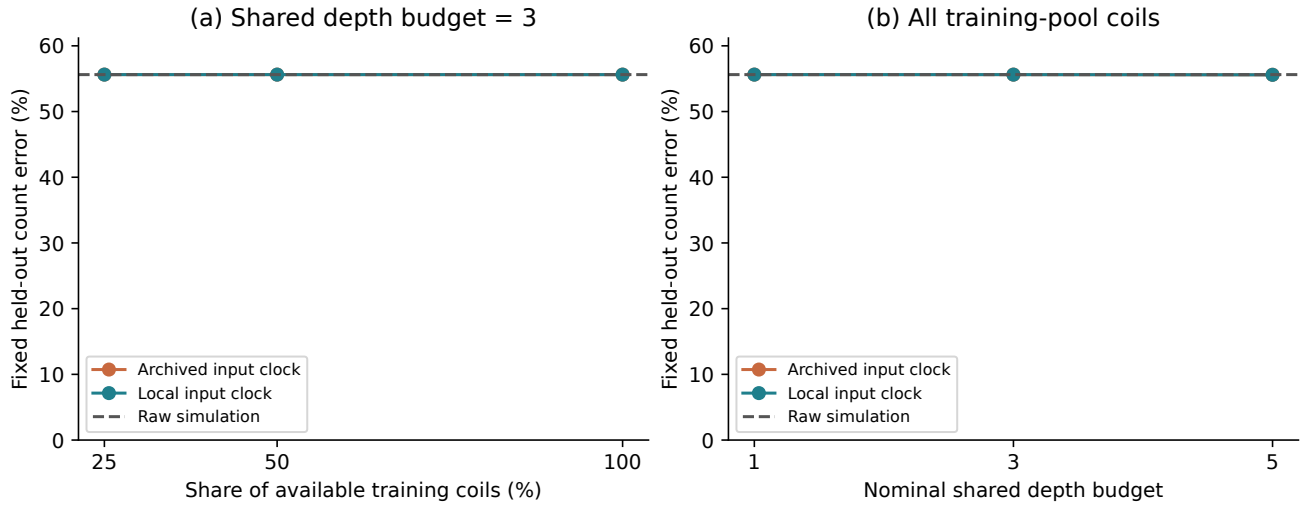


Figure S6: Exploratory fixed-holdout experiment on one date. (a) Training-pool coverage at shared recursion budget 3; shading is the range across three nested subset repeats, not a confidence interval. (b) Nominal depth budget with all training-pool coils. All evaluation targets are in local time, and the raw error is dashed. The near-overlapping curves reflect negligible gains on this particular withheld set.

S7 Reproducibility and unresolved experiments

From the parent repository root, run `python3 article1/analysis/validate_archive.py`. This reads the original simulations, GPS extracts, MTS files and archived held-out counter table. It creates aggregate JSON/CSV results, the two new vector figures and the main baseline table under `article1/`. Link-level GPS-derived outputs remain under the ignored `results/private/` directory. SHA-256 digests of

simulation, configuration and GPS inputs and software versions are recorded in the validation summary. Counter source digests are listed in the package inventory.

The existing counter predictions were originally produced by `article/python/revision/loo/run_loo.py`, using the historic correction code and metadata recovered from project history. The new script resamples these predictions; it does not rerun the counter experiment. Historic scripts are referenced for provenance and must not be executed unchanged if preserving the original output directory is required.

The bibliography metadata is checked with `python3 article1/analysis/verify_references.py`; saved Crossref responses permit inspection without relying on manually transcribed titles or DOIs. Network failures are recorded and prevent replacement of the bibliography. OpenStreetMap is cited as the network source, not as a Crossref-indexed publication.

The current assessment does not include full-campaign ensembles for OD assumptions, sensor coverage, graph depth or random seed. Corrected-demand configurations in the original rerun workspace are candidates for sensitivity experiments; their hour-label interpretation still requires confirmation. A one-day replay completed in 607 s with matching keys but failed numeric reproduction: summed link-hour counts are 33,361,561 against 34,334,965 in the archive. Its cause has not been isolated. A fixed seed does not establish historical reproducibility. These tasks and their current status are maintained in `article1/PLAN.md`.

All maps and composite figures inherited from the source draft are marked as such in the main captions. The validation, diagnostics and supplementary sensor-pilot figures have been generated by the new pipelines. Road-class stratification is deferred until a reliable classification is joined to the model graph. No proprietary data or manuscript has been uploaded or published.